



ESCUELA DE INGENIERÍA EN COMPUTACIÓN
PROGRAMA DE MAESTRÍA EN COMPUTACIÓN

Thesis Title Goes Here

A thesis submitted in partial fulfilment to opt for the degree of

Magister Scientiæ in Computer Science

Author
Student Name Here

Supervisor
Supervisor Name Here, Sup. Title

September 16, 2026

Thesis approval certificate goes in here.

If the author wants, he can put a dedication here.

Y la versión en español puede ir acá.

Acknowledgements

This section can be used for the acknowledgements.

Abstract

This document presents the thesis topic proposal for obtaining the Master's degree in Computer Science: Emphasis in Computer Science at the Tecnológico de Costa Rica. The proposed topic is centered in the idea of the potential use of machine learning techniques to analyze the patterns in acoustic echoes produced inside small rooms to estimate the location of certain objects in a tridimensional space. Two neural networks architectures are proposed for this task, a Convolutional Neural Networks that despite being an architecture mainly for images is commonly used in literature for acoustic inference, and an Echo State Machine as a new architecture with potential to address directly the acoustic signals. The effect of beamforming is also considered as one of the elements to investigate as a solution to the multi-path acoustic superposition. The design for the research is presented, the required experimental setup is discussed as well as the methodology to measure, process and evaluate the results, all this with the objective to evaluate the potential use of sound echoes in the audible frequency as a mean to locate object in an enclosed room.

Contents

1. Introduction	1
1.1. Problem	3
1.2. Document Structure	4
2. Theoretical Framework	5
2.1. Room Impulse Response	5
2.2. Acoustic Beamforming	6
2.2.1. Microphone Array	7
2.2.2. Beamforming Pipe	8
2.2.2.1. Delay-and-Sum Beamforming Technique	8
2.2.2.2. Deconvolution	8
2.2.2.3. Post-Filtering	9
2.3. Time Series Transformation into Images	9
2.3.1. Markov transition Field	9
2.3.2. Gramian Angular Field	11
2.4. Convolutional Neural Network	11

2.4.1.	Convolution operation	11
2.4.2.	Pooling	13
2.4.3.	Fully connected layer	14
2.5.	Echo State Network	14
2.5.1.	Formalism of ESN	15
2.6.	Related Work	16
2.6.1.	Passive Acoustic Sensing and Analysis	17
2.6.1.1.	Acoustic Imaging	17
2.6.1.2.	Acoustic Scene Classification	17
2.6.1.3.	Acoustic Object Recognition	19
2.6.2.	Active Acoustic Sensing and Analysis	20
2.6.2.1.	RIR for Room Inference	20
2.6.2.2.	Acoustic Object Shape Estimation	21
3.	Hypothesis and Objectives	23
3.1.	Hypothesis	23
3.2.	Objectives	24
3.2.1.	Main Objective	24
3.2.2.	Specific Objectives	24
3.2.3.	Deliverables	25
3.3.	Scope and Limitations	25
4.	Methodology	27
4.1.	Experiment Design	27
4.2.	Experimental measurement configuration	28
4.3.	The Audio-Location Dataset for Model Training and Testing	29
4.4.	The Audio-Location Processing Pipeline	31
4.5.	Factors and Levels	33

4.6. Response Variables	34
4.7. Data Recollection for Model Evaluation	34
4.8. Statistical Tool Used for hipotesis Testing	35
5. Design	37
5.1. Design criteria used (if needed)	38
5.1.1. Detailed specific criteria/tool	38
5.2. Description of the implementation of other algorithms (by other authors) used to test the hypothesis	38
5.3. Proposed algorithm	38
5.3.1. Design of the proposed algorithm	38
5.3.2. Limitations and Requirements	38
5.3.3. Implementation	38
6. Results	41
6.1. Performance evaluation	42
6.2. Statistical Tool Assumptions Tests	42
6.3. Statistical Tool Results	42
6.4. Obtained Metrics	42
6.4.1. Most Complex Scenario Metrics (if needed)	42
6.5. Correctness (if needed)	42
6.6. Output (If needed)	43
7. Discussion	45
8. Conclusions and Future Work	47
8.1. Conclusions	48
8.2. Future Work	48

9. Appendixes	49
9.1. Source Code	50
9.2. Gathered Data from the Experiments	50
References	51

List of Figures

2.1. Room Impulse Response and the geometrical room representation [28] .	6
2.2. Microphone array in binaural arrangement [38]	7
2.3. Summarized process to build a Markov Transition Field [10]	10
2.4. Convolution operation over a tensor of 5x5 with zero padding and a stride of 1, resulting in a feature map of 5x5 [42]	12
2.5. Pooling escalation with average pooling and max-pooling	13
2.6. Structure of Convolutional Neural Network for aircraft structural health monitoring [33]	14
2.7. Comparison of B) RC vrs A) traditional RNN, the bold arrows are weights that change during training [20]	15
2.8. Image on the left is an acoustic image, on the right and microphone camera used to capture the image [29]	17
4.1. Measure box: (a) Conceptual sketch for the acoustic measurement box, and (b) actual implementation for the data acquisition task.	29

4.2.	Scatter plot of cube locations for each recording in the dataset, the blue columns represent the 26 sessions, and each point in the column represents the nine orientations captured at that location.	31
4.3.	Acoustic to location processing pipeline	32
5.1.	Example diagram.	39

List of Tables

4.1. Factors and levels for the experiment.	33
4.2. Five-fold cross-validation structure and performance summary.	35

Introduction

Robots are expected to navigate in multiples and different environments, in order to achieve this we need to allow them to have situational awareness and autonomy to make proper decisions. Computer vision aims to enable machines to get an interpretation of the world that surround them. Processing images and videos robots can extract information to recognize objects geometries, textures, materials, etc. Although it is true that with visual analysis, the machine can create internal models that contain a vast knowledge of the environment, this doesn't mean that its perception is complete. Robotic perception give the ability to a machine to understand its surroundings to the degree that could safely interact with the world [12], for such understanding other techniques are required as well as computer vision to get a complete sensory experience of the environment. Acoustic sensing and analysis can enhance the perception of machinery following an approach similar to that seen in nature with animals that use binaural acoustic perception, the environment can be analyzed to detect the source of a particular sound and focus the visual attention to that particular region, but, this capability is not only an extension,

in some conditions the animal can entirely relies on the acoustic information to operate, this is the case of echolocation employed by bats, birds, dolphins and wales, they ear the echoes of acoustic chirps to gain spatial perception of the surrounding environment and this enable them to navigate in complex layouts [40].

Traditional computer vision techniques that employ sensing devices such as cameras and lidars provide high-fidelity information to navigation systems, but they become unreliable in scenarios with poor light or presence dust, smoke, and fog [34]. Another aspect to consider is the cost of the sensing device, here the acoustic perception offers an advantage due the low cost of commercial electret or MEMS microphones which are inexpensive compared to optical cameras and lidars; then become desirable its use as an alternative to traditional sensors. Recent developments in machine learning as the ones shows in [2; 27; 34] demonstrated the big potential to use neural network to infer spatial and geometrical information of objects in the surrounds by just analyzing acoustic echoes in the audible spectrum. The manufacturing sector under the industry 4.0 transformation is increasing in the adoption of automation and smart monitoring technologies and so it has become an important field of application of these new methods, where the supervision of machinery during operation, especially the machines allocated in production floors using robotic arms and computer numerical control actuators intended to manipulate objects in confined spaces [24]. In such conditions the acoustic sensing is a good approach to get an insight of the overall status of the system due to the nature of the Room Impulse Response as a transfer function that comprehends information of the environment setup. The acoustic response of a room can be use together with artificial neural networks like convolucional neural networks or recurrent neural networks to learn from this response and be capable of recognize characteristics and properties of the room echoes to recognize spaces and objects in it. Thus, acoustic perception together with neural networks can improve the perception of machinery

when it operates in complex spaces.

1.1. Problem

The use of acoustic signal to approximate the size and shape of objects was considerably improved by the adoption of machine learning techniques. An important part of the work in this field was develop with the scope of indoor navigation systems where the objective is to enhance the perception of robots using acoustic analysis to build low resolution deep images of an unknown environment. A different point of view is formulated and presented in this work, one in which we know the environment completely a priori, but we have an object that will change it's position inside the environment and we expect to know it's location with certain accuracy each time. Second, contrary to the large size room in which those systems are intended to navigate, we expect to estimate the object location in short ranges in which multi-path echoes could be received at the same time creating a superposition difficult to resolve and cause malfunction in those systems [2]. It is our understanding that there are no works related to estimating the position of an object through acoustic analysis under the circumstances described. To achieve the acoustic spatial estimation in this scenario we propose two key steps to the processing pipeline, one in the beamforming stage to increase the spatial discrimination from where the echoes are originated, and second a change in the type of neural network use to infer the location, to employ a network suitable to directly process time series instead of Convolutional Neural Networks. Our purpose is to demonstrate that is possible to locate an object in a small room (e.g. 1m side dimension) using acoustic signal processing with neural networks.

1.2. Document Structure

This document present in **Chapter 1** a brief introduction to the topic and is stated the problem we intent to address with the development of this thesis. **Chapter 2** contains the theoretical background and state of the art in the field. **Chapter 3** details the hypothesis, objectives and scope of the work. **Chapter 4** shows the research proposal and methodology. **Chapter 5** present the schedule for the execution of the proposed work.

Theoretical Framework

In this chapter is described the theoretical concepts needed to understand the work that will be developed during the research's execution.

2.1. Room Impulse Response

An acoustic impulse is sharp and narrow pulse which spectrum is composed by a uniform distribution of all frequencies. The mathematics Dirac delta function $\delta(t)$ is used to represent this kind of impulse. In enclosed spaces the sound waves generated by an impulse will travel and reflect and refract in the wall delimiting the space, for a microphone measuring the environment, it will receive the direct path sound, the sound that travel from the source directly to the microphone, followed by the early reflections and finally a series of pulses called late field reverberation as show in figure 2.1 . The direct path, early reflection and reverberations correspond to the response of the enclosed space to an acoustic impulse and is physically defined by the environmental geometry and materials [31].

The Room Impulse Response (RIR) by definition is the transfer function of a sound wave that travel in the space from it's source to the microphone. The microphones hear the convolution of the emitted sound with the corresponding RIR [8] mathematically showed in equation 1, therefore any sound generated in a point and received in other inside a room is the composition of the original signal plus the RIR of the enviroment.

$$g(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(t)x(t-s)ds \quad (1)$$

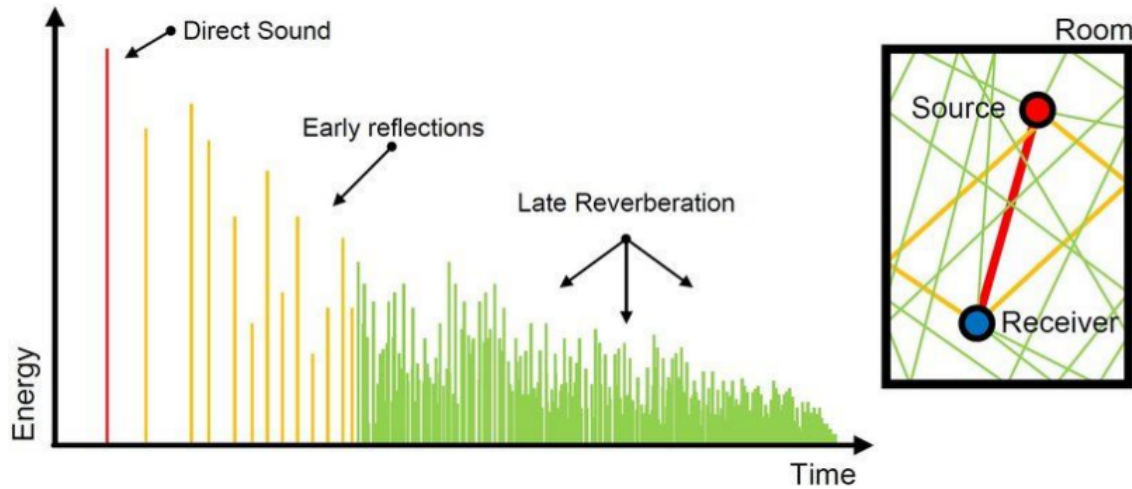


Figure 2.1: Room Impulse Response and the geometrical room representation [28]

2.2. Acoustic Beamforming

In many spatial omnidirectional data transfer context, not only the information carried in the medium is important, the source from where the signal is generated also provide valuable information to properly analyze the incoming data. Beamforming is a spatial filtering technique aims to estimate the direction of the arrival signal inclusive in the presence of noise and other interference in the propagation medium [5]. In acoustic

an array of microphones is used to capture the spatially propagated signals arriving from a certain direction and processes them to investigate the acoustic sources [23].

2.2.1. Microphone Array

A Microphone array combine the signal of multiple omnidirectional microphones in such a way that capture sound from a specific direction while reject sound from other directions [38]. The shape and dimensionality of arrangement depend on the application. The simplest form consist in two microphones spaced by a distance D as show in figure 2.2. The value for D is the minimum distance for a given frequency and its calculation is shown in the equation 2, in the case that the input signals are simple added together for a given frequency f , this arrangement capture sound coming from ± 90 degrees and rejects the sound at 0 and 180 degrees. In a microphone array the higher the number of microphones results in better gain in the desired direction as well as the attenuation of sound coming from the sides of the array [30].

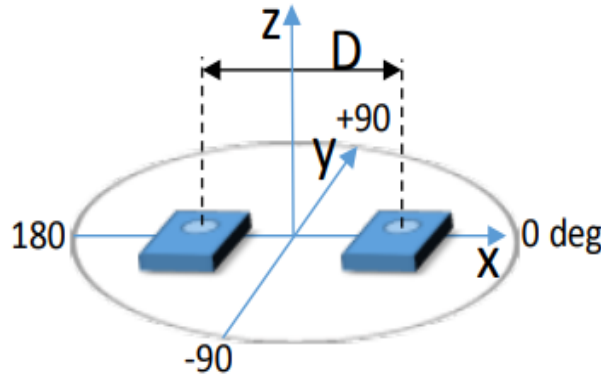


Figure 2.2: Microphone array in binaural arrangement [38]

$$D = \frac{1}{4} \frac{c}{f} \quad (2)$$

2.2.2. Beamforming Pipe

2.2.2.1. Delay-and-Sum Beamforming Technique

Beamforming is based on the physical principle that an acoustic wave originated on a particular location in the space, it travels and arrive at each sensor at different times defined by the distance existing between the unique source and each sensor. The result of adding all these signals depends of the phase relationship between them, so is possible to control the delay for each signal to constructive add the ones from a particular location and, at the same time, destructively cancel the ones originated in other regions [5]. Mathematically the beamforming formula is show in equation 3, where $L(t, S_0)$ is the beamforming output, pm is the microphone output for a signal originated at $S_0 = (x_0, y_0, z_0)$ and captured by a microphone located at S . All the signals are sum together and divided by the total number of microphones M in the array.

$$L(t, s_0) = \frac{4\pi}{M} \sum_{m=1}^M pm(s_0, t + t_0) |S - S_0| \quad (3)$$

2.2.2.2. Deconvolution

In real applications the beamforming output comprehends the acoustic signal convolved with the microphone array response. The deconvolution principle assumes that the recorded signal $r(t)$ is a convolution of the sound $g(t)$ (The result of equation 1) and a transfer function $ma(t)$ which represents the array response to a given source [5]. The Point-Spread-Function (PSF) is the array response to a monopole placed in front of the array. The PSF provides the information required to obtain the array transfer function required to do deconvolution and extract the sound without the distortion created by the array.

2.2.2.3. Post-Filtering

Taking into consideration that the signal measured for each microphone is the sum of the sound and the level of noise, and also that the noises are uncorrelated between each microphone and between the beamforming output signal. The beamforming output can be further enhanced by selectively filter the resulting signal based on the cross-spectral densities between channels in the microphone array [23].

2.3. Time Series Transformation into Images

Given a time series, for the sake of analysis sometimes is desirable to perform certain kind of transformations to the time series that provide a more compact representation of the data or highlight specif properties and attributes of it. Transforming time series into images involves the process of transforming vectors into matrices using a criterion of transformation that map values from one domain to another.

2.3.1. Markov transition Field

In the Markov trasnsition fields (MTF), given a time series $X = (x_1, \dots, x_n)$ of real observations, the serie is discretized based on quantile bins assignation. First the codomine of the function is sub-grouped taking the range of values in X and divided it into Q segments, then each x_i is allocated to bin q_j with $j \in \{1, \dots, Q\}$ if its value is within the segment. Now considering the resulting bin values as observations of a first-order Markov chain and computing the number of occurrences of pairs of back-to-back bins for every pair of bins is possible to represent the ocurrences as a Markov transition matrix of dimension $Q \times Q$. The Markov transition matrix is normalized to transform frequencies into probabilities. then each entry in the matrix is the probability to go from q_j to q_k , for every pair (q_j, q_k) [10].

Finally the dimensionality of the time series is restored by projecting the Markov

matrix $Q \times Q$ into a $n \times n$ matrix called Markov transition field, where each entry for the pair (x_i, x_j) is the probability to go from the bin $q(x_i)$ to bin $q(x_j)$. For a serie of n observations the MTF can formally be defined by the equation 4.

$$\forall i, j \in 1, \dots, n, MTF_{i,j} = \mathbb{P}(q(x_i)|q(x_j)) \quad (4)$$

The figure 2.3 Shows the data trasformations requiered to obtain a MTF, First in (a) the time series is discretized by dividing the variable range in intervals, (b) the Markov transition matrix is calculated and from this the MTF for the probability of each pair in the serie is created as presented in (c).

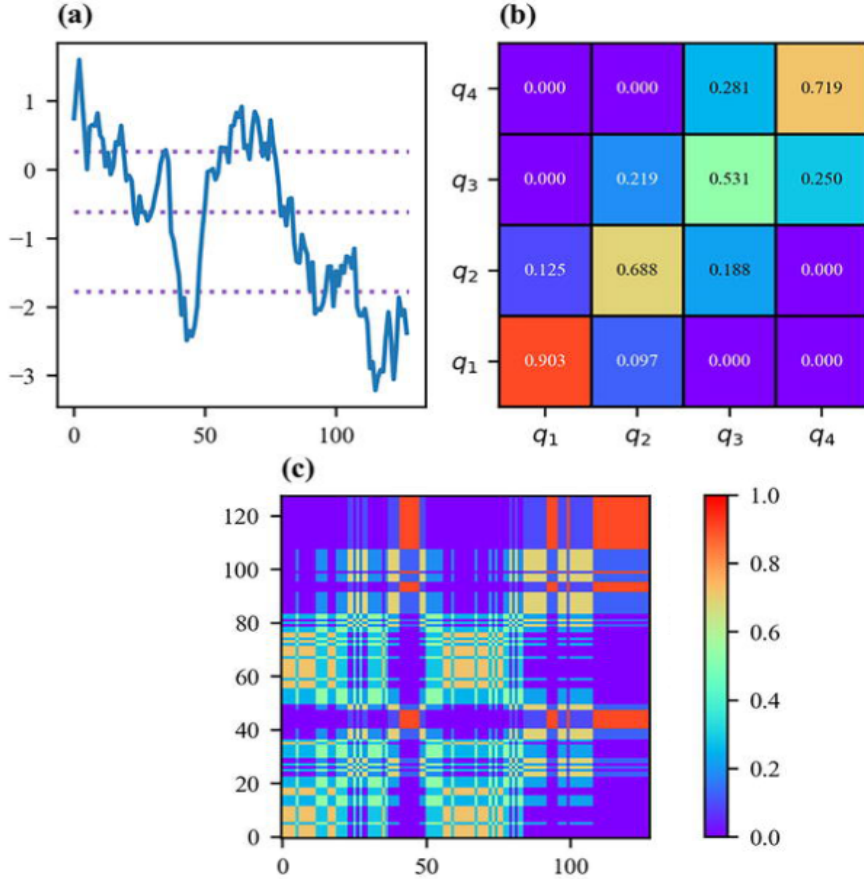


Figure 2.3: Summarized process to build a Markov Transition Field [10]

2.3.2. Gramian Angular Field

The Gramian Angular Field (GAF) is a matrix build on the principle of transform the time series values to the polar domain [37]. Given a time series $X = (x_1, \dots, x_n)$ of real observations, the gramian transformation takes the values x_i and rescale them in the interval $[-1, 1]$ or $[0, 1]$. Then the rescaled values are converted to a polar form considering the rescaled value as the angular cosine and the timestamp the radius. The transformation function is show in 4. From the resulting values only the angle is used as the one that depends of the values of the time series. Two matrices can be build considering the sum or subtraction of pair of angles resulting from 4, the sum is called Gramian Angular Sum Field (GASF), the subtraction of angles receive the name of Gramian Angular Difference Field [11].

$$\begin{aligned}\phi_i &= \arccos(\bar{x}_i), \quad \bar{x}_i \in \bar{X} \\ r_i &= \frac{t_i}{n}, \quad t_i \in \mathbb{N}\end{aligned}\tag{4}$$

2.4. Convolutional Neural Network

A Convolutional Neural Network (CNN) is an artificial neural network architecture used to analyze and learn from visual data presented commonly in the form of images. In its simplest form the structure is composed by three building blocks: convolution layer, pooling layer, and fully connected layer. The figure 2.6 shows and example of how this blocks are connected to analyze an image stage by stages in order to determine the structural condition of an aircraft fuselage.

2.4.1. Convolution operation

Convolution is a common operation used on image filtering to denoise or detect features. A filter also called a convolution kernel is applied to all locations of the image to produce a new filtered image [7]. For a given 2D image X of dimension $I \times J$ with

pixels $x_{i,j}$ a filter W of dimension $K \times L$ with weights $w_{k,l}$ can be convolved to obtain a filtered image $X * W$. The convolution process over the image can be mathematically written as:

$$(w * x)_{i,j} = \sum_{k=-K/2}^{K/2} \sum_{l=-L/2}^{L/2} w_{k,l} x_{i+k,j+l} \quad (5)$$

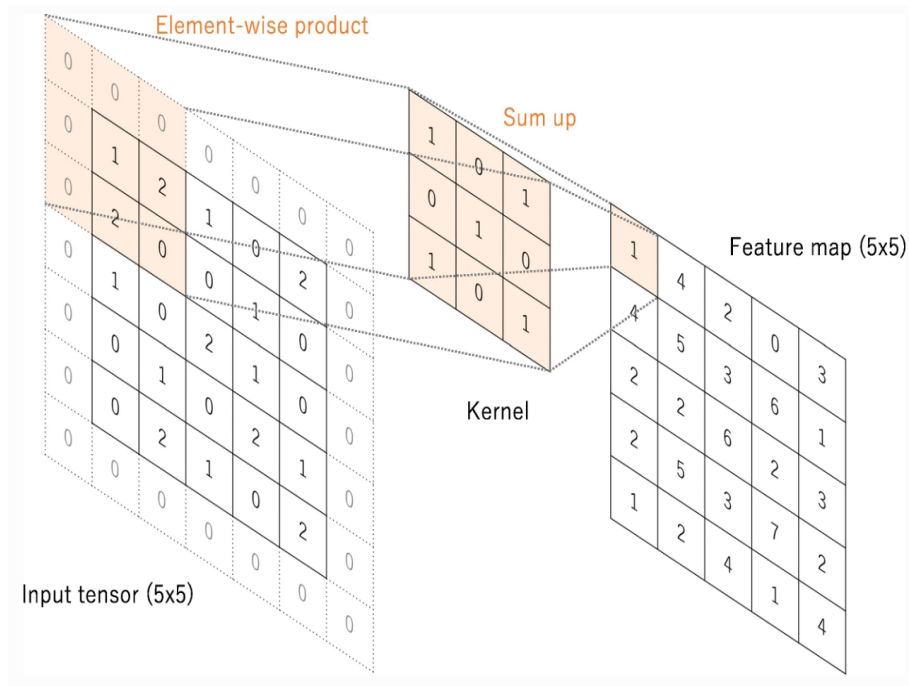


Figure 2.4: Convolution operation over a tensor of 5x5 with zero padding and a stride of 1, resulting in a feature map of 5x5 [42]

The convolution operation in equation 5 does not allow the center of the kernel to overlap the edges of the image, resulting in a reduction in the size of the output feature map. Padding increases the input size by adding extra rows and columns around the outermost elements on each side of the image in order to compensate the dimensional reduction [42]. In practice as shown in figure 2.4 the kernel is applied across the input one step at a time, an element-wise operation is performed in this

location and summed to obtain the output, after this the kernel is moved to the next location to filter other segment of the input. How big is displacement of the kernel is given by the stride, which define the distance between two successive kernel positions.

2.4.2. Pooling

Pooling is a dimension reduction technique where an input map is spatially reduced while keeping the most important features filtered or learned. The pooling operation reduce or collapse a sub region to a smaller dimension helping the learning representation to be more invariant to small length-scale perturbations [16].

Two common pooling techniques are max-pooling and average-pooling, the first one, for the defined pooling window it will consider only the maximum value, in the case of the average pooling is calculated the average of the values inside the window as show in the example of figure 2.5.

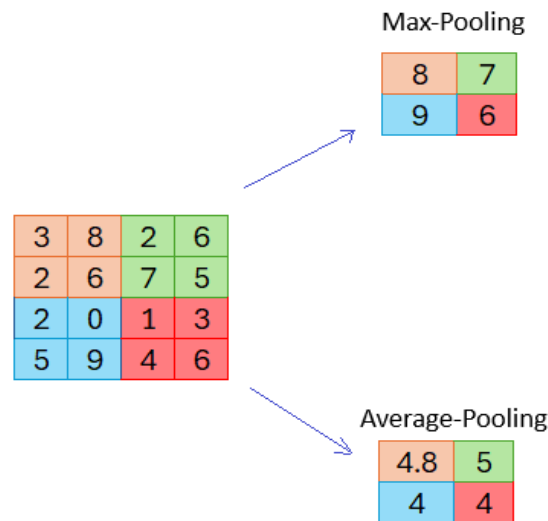


Figure 2.5: Pooling escalation with average pooling and max-pooling

2.4.3. Fully connected layer

The final stage in a CNN is a fully connected layer. This layer is responsible to classify the features extracted from the previous layers. The figure 2.6 shows an example of CNN for aircraft structural health classification with its fully connected layer allocated at the output of the net.

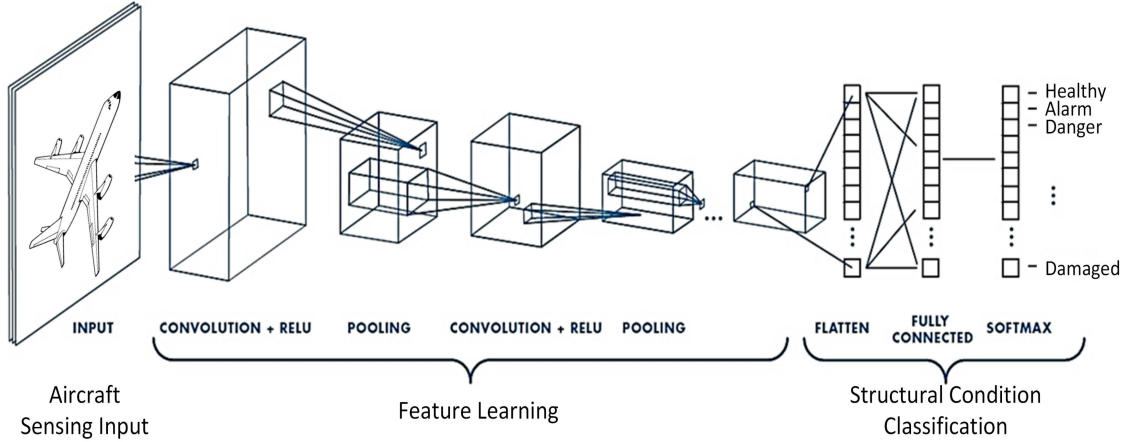


Figure 2.6: Structure of Convolutional Neural Network for aircraft structural health monitoring [33]

2.5. Echo State Network

An Echo State Network (ESN) is a new paradigm in recurrent neural networks called Reservoir Computing (RC). In RC, the reservoir is the recurrent part of the network and has the particularity that it is not trained. Due to the fact that the reservoir skips the training, the network avoids the computationally expensive gradient-descent on recurrent networks, and instead of training the recurrent network, it is defined in such a way that it is capable of transforming input signals into a high-dimensional state space. The state of the reservoir can be processed by a simple fully connected readout network trained to map the reservoir state to a desired output [20].

In the ESN, the reservoir is a randomly created RNN where the weights remain

unchanged. This RNN is exited by an input signal, and its future state is a function of the actual input and the history of the reservoir. The result is generated by training a simple feed-forward network to output the target value. Figure 2.7 [20] shows a comparison of traditional RNN and RC. In RC the bold arrows used to exemplify the weight change resulting from the training process and is only performed at the output layer of the network.

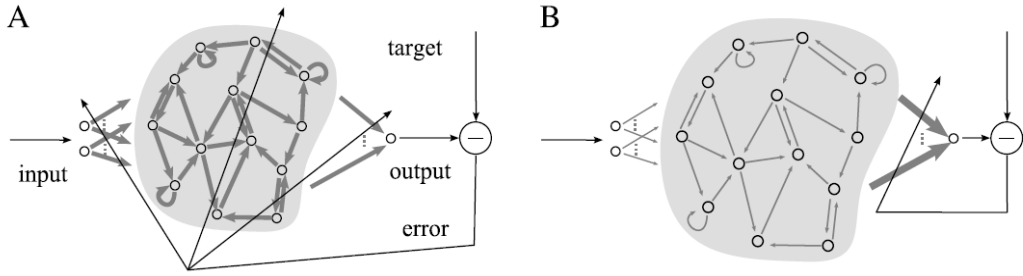


Figure 2.7: Comparison of B) RC vs A) traditional RNN, the bold arrows are weights that change during training [20]

2.5.1. Formalism of ESN

A formal description of an ESN from [32] define an ESN receiving a time series input $U(t) \in R^{D \times 1}$, the state of the reservoir is denoted by $X(t) \in R^{N \times 1}$ and $y(t) \in R^{M \times 1}$ the output. The weights for the stages are: $W_{in} \in R^{N \times D}$ for the input, $W_{res} \in R^{N \times N}$ for the reservoir and $W_{out} \in R^{N \times M}$ for the output. The equations for the internal state of the reservoir and the output are show in 6. The reservoir state depends of the input value at time t and the previous state of the system. The output can be obtained by training to solve a linear regression problem.

$$\begin{aligned}
 x(t) &= f(W_{in}U(t) + W_{res}x(t-1)) \\
 y(t) &= W_{out}x(t)
 \end{aligned}
 \tag{6}$$

There exists an important property that the reservoir should meet to ensure the reservoir dynamics is capable of forget the initial transient state [41]. The Echo State Property (ESP) enunciate that a network with state transition equation F has the echo state property if for each input sequence $U = [u(1), \dots, u(N)]$ and all couples of initial states x, x' for the reservoir, it hold the equation 7.

$$||F(U, x) - F(U, x')|| \rightarrow 0 \text{ as } N \rightarrow \infty \quad (7)$$

To ensure the existence of ESP the W_{res} must meet a necessary condition and a sufficient condition as defined as follow:

$$\begin{aligned} ESP &\Rightarrow \rho(W_{res}) < 1 \\ \bar{\sigma}(W_{res}) < 1 &\Rightarrow ESP \end{aligned} \quad (8)$$

Where $\rho(W_{res})$ is the spectral radius parameter of W_{res} and the $\bar{\sigma}(W_{res})$ is the largest singular value of W_{res} . So from equations 8, if $\rho(W_{res})$ and $\bar{\sigma}(W_{res})$ are less than one, the ESP is satisfied and the reservoir dynamic will be correct.

2.6. Related Work

The related work in acoustic processing for object recognition and classification can be divided in two categories based on from where the sound is originated. We can considered a passive sensing approach if the sound stimulus is originated directly in the system, otherwise if the sound is an external stimulus that interact with the

environment before sensed we are under a active sensing approach.

2.6.1. Passive Acoustic Sensing and Analysis

2.6.1.1. Acoustic Imaging

An acoustic image is a sound intensity map that highlight the location of different sound sources in a bi-dimensional plane. An acoustic camera takes this intensity map and superimposed it on a optical image to show the physical source of the sound. As show in figure 2.8 a microphone array is used to ear the sound and a beamforming algorithm process the signals to locate the direction from where the sound is originated.

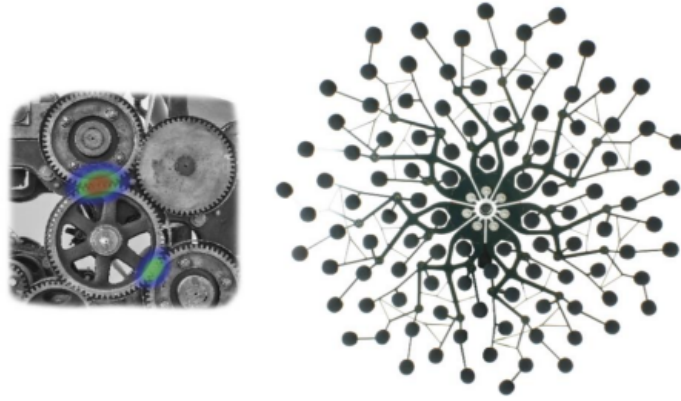


Figure 2.8: Image on the left is an acoustic image, on the right and microphone camera used to capture the image [29]

Acoustic beamforming is a field that has been extensively studied, different shapes and sizes of microphone arrays has been analyzed [30], also exist many techniques to beamforming the signals from the array [23], recently studies are focused on spherical arrays to identify multiple sound sources in a three dimensional space [6; 14; 18].

2.6.1.2. Acoustic Scene Classification

Artificial Intelligence and specially the deep learning subfield powered the development of new techniques to segment, analyze and classify acoustic originated

signals. Acoustic Scene Classification (ASC) use deep learning to learn and distinguish from audio the context in which the recording was performed. The CNN are the first choice for ASC, the required images that are going to be the input to the network are obtained by processing an audio dataset to convert it to spectrograms. Taking the DCASE 2016 dataset, the static Log-Mel spectrograms and dynamic first derivative spectrograms are used in [4] to build the two channels of the images that will be used as input training of the CNN. Han et al. in [13] used the DCASE 2017 dataset to improve the input data to train, this through multiple versions of mel-spectrograms divided in binaural representation, source separation and background subtraction. In binaural representation instead of mono audio representation they used left-right and mid-side spectrograms. The source separations divide the dataset into two using harmonic-persecutive sound separation algorithms to increase the dataset to get the representation in armonic and percusive sound. Finally media filtering is used in the spectrograms to steady noise from the environment. The resulting multiple version spectrograms are used to training a ConvNet architecture of 8 convolution layers inspired in a VGG net. They obtained a mean accuracy improvement compared with the baseline used in the work and a final 74.8% to a 80.5% accuracy with the various prepossessing methods previously described. The work presented in [36] used the DCASE 2019 dataset and the log-mel spectrograms to training two deep learning algorithms to perform the classification, a CNN but also a Convolutional Recurrent Neural Network (CRNN) are the architectures choosed in this work. The highest recognition obtained was for the CRNN with 90.96% of accuracy, a value superior to 73.61% of the CNN used as baseline. A different objective is presented by Byttebier in [3] where small foodprint neural network models are investigated. The author compared several small neural networks employing the DCASE 2020 to evaluate them and found that a 16-bit CNN with squeeze excitation blocks, regularized with MIX UP and weight decay is the one that shows the best

accuracy of 66.84% with only 90 kilobyte of parameters.

2.6.1.3. Acoustic Object Recognition

Acoustic object recognition aims to recognize an object in base of the sound that emits whether it is self-generated as the sound produced by and air-conditioner during normal operation or external, in response to a stimulus like the characteristic sound produced by the wood when someone is knocking a door. For the work presented by Luo et al. in [21] a dataset was built using 30 different objects, each of them was hit by a marker pen to generate a characteristic sound that was recorded. A training set of 100 hits (in different locations of the object) per object was created, and an extra 20 hits per object were obtained as well in order to be use in the test phase. With the dataset a denoise-autoencoder was trained to recognize the different sound of the object from which its originated. The recognition rate for the autoendoders was 91.50%, a value superior to the baseline provided by a Support Vector Machine with recognition rate of 82%.

Other dataset was created by Tran et al. in [35], in this case the input were LEGO objects that fall from 30cm high. The sound of the LEGO pieces hitting the ground was used to create spectrograms to training a CNN called LegoNet in order to recognize the LEGO pieces by the sound of its free-fall impact, LegoNet achieved an accuracy of 99.21%, a value superior to the other networks used for comparison. For robotic perception in [17] was develop a framework in which a robot can recognize the object inside a container by shaking the container and analyzing the sound. An kernel K-nearest neighbor algorithm in a open environment is used to shows the feasibility of the method. A generative physical, audio-visual synthesis pipeline is presented in [43], the model takes as input the shape of an object, its material and one scene configuration, a physical engine simulate motion and collision for that particular object. An audio engine takes the shape, material and the collision information to synthesize the sound.

The database generated is used in the training a two-stream CNN, one ResNet-18 for the image data, together with a SoundNet-8 for the audio data. The net demonstrated that the knowledge learned in the synthetic audio-visual dataset was transferred to real world data once it was used to infer object materials from auditory data, and estimate shape attributes from audio-visual data.

2.6.2. Active Acoustic Sensing and Analysis

2.6.2.1. RIR for Room Inference

For an enclosed space, the RIR provided a unique representation of the geometry and material inside it, sound like music and speech recorded in rooms comprehend this echoes information and can be use to recognize the room in which the recording was performed. This idea is presented in [25] where the authors developed a machine learning system to identify rooms for the reverberant audio samples recorded in those rooms and achieved and 85% accuracy in the task. The work in [1] use the RIR and speech data to train a model to infer the room volume from a single microphone recorded sound, with this information the model with an accuracy of 85.8% classified the rooms into three categories: small, large or hall. Using microphone arrays and analyzing the RIR obtained from them in [9] the authors are able to estimate the room geometry by exploring the properties of Euclidean distance matrices and show that under certain conditions the first order echoes provide a unique description of the geometry of convex polyhedral rooms. In [19] an indoor positioning device is presented using the RIR as the "fingerprint" to identify the room in which the system is located, the accuracy varies depending of the type of room, ranging from 65% in the identification of living rooms to 95% respect to bathrooms. The work in [39] shows a dataset of RIR in which people was standing in the rooms and relocated to different positions inside it to generate samples. The author shows the potential of using machine learning approaches over the dataset for tasks related to localizing, identifying and detect humans in those rooms.

2.6.2.2. Acoustic Object Shape Estimation

Acoustic perception of shapes are explored in [22], this work proposes the use of acoustic standing waves to recognize the size and shape of a target object analyzing the distance spectrum profile of the standing waves created by the interference between transmitted and reflected sounds, In this work Manabe et al. were capable of estimate the dimensions of a rectangular board of 1.2m x 0.9m and the mean absolute error of the estimation was 0.0194m, with this result they demonstrated that the spectrum profile encodes information of the target and can be use to estimate the shape of an object. The approach presented by Brunetto et al. in [2] result in the developed of the Batvision system capable of see deep spacial layouts by listening to echos recorded by two microphones as artificial ears. For Batvision a dataset is created using mobile robots that generate acoustic chirps signals at the same time a deep image is captured, with this dataset of pairs audio-image an adversarial discriminator is trained to generate deep images based on sound chirps. The system was capable of predict detailed indoor scenes and obstacles with a L1 loss as low as 0.072, sometimes even outperforming a deep mapping created by a stereo vision algorithm. Following the findings in Batvision the work presented in CatChatter paper [34] shows two separate results, first an improved version of Batvision is developed in which an input encoder from the model was removed, this encoder was processing the acoustic spectrogram before reaching the adversarial discriminator and was causing that spatial inputs from the spectrogram were lost at the time it reach to the U-Net decreasing the inference capacity of the model. The other result presented in [34] was from a new system developed that use lidar to create 3D, 360 degrees depth images together with an increase in the acoustic perception from 2 microphones to 4 in order to capture echoes from all the directions. The results obtained by this new method are similar to the ones reported in Batvision [2] but the field of prediction due to 360 deep information and the extra 2 microphones was 4 times bigger than the obtained with deep camera solution and two microphones system. To detect

basic shapes and deformation acoustic sensing was used in [27] by Rothstein et al. to identify and monitor small hollow object from inside. The deformation of the shapes are created by generating similar shapes with alter dimensions or by blocking a certain part of the object interior. A square wave sound signal is emitted in the interior of the objects and a 4 microphone array records the echos produced in the interior. Acoustic features are created from these signals and a classifier is trained to distinguish the shape and deformation of the object under study. Their work presented that for the objects the shape can be identify with 80% accuracy, the existence of deformation with 100% accuracy and localize the deformation with 90% accuracy.

Hypothesis and Objectives

3.1. Hypothesis

The compendium of work presented in the Chapter 2 demonstrated that is possible detect with some precision the source of a sound in a tridimensional space, recognize a particular room by its unique acoustic and even create a deep image of segments of this room using only audible sound echoes and infer shapes of object based on acoustic features. Those results support the idea that for a given object inside a enclosed space is possible to learn a function that map the RIR to the location of the object in that space. Based on these ideas our hypothesis is:

"An algorithm pipeline with beamforming and ESN that utilize the rich spatial-geometrical information comprehended in a set of RIR of acoustic signals can infer the location and orientation of a cube of 20cm side allocated inside an small enclosed space of near of $1m^3$ ¹, this with a movement distance resolution of at least 5cm and achieve

¹Region with acoustic superposition found by [2]

less than 2cm of error in position and 10 degrees in orientation² with an error less than 6 degrees in the estimation from two consecutive positions."

3.2. Objectives

The current research established the following objectives:

3.2.1. Main Objective

To model and evaluate a processing pipeline using beamforming together with CNN and ESN to transform a multichannel RIR acoustic signals to spatial coordinates and angles that situate an object within an enclosed space.

3.2.2. Specific Objectives

1. Collect a dataset of acoustic responses of an object positioned in different locations inside a room.
2. Develop a reference method using CNN to learn from the dataset a function that estimate the coordinates and orientation of an object inside the space.
3. Develop a resources efficient method using ESN to learn from the dataset a function that estimate the coordinates and orientation of an object inside the space.
4. Build a beamforming stage that divide the measure space in four regions of acoustic sources.
5. Evaluate experimentally the accuracy of the estimators.

²Based on the size of the objects and maximin deformation presented in [27]

3.2.3. Deliverables

Specific Objective 1: A dataset will be delivered containing acoustic recordings of echoes in the form of time series and its corresponding object location-orientation.

Specific Objective 2: Thesis document with detailed description of the concepts, observation and results analysis that justify the system design with CNN.

Specific Objective 3: Thesis document with detailed description of the concepts, observation and results analysis that justify the system design with ESN.

Specific Objective 4: Thesis document with detailed description of the concepts, observation and results analysis that justify the beamforming.

Specific Objective 5: Thesis document with analysis and evaluation of the proposed solutions considering performance metrics under experimental operation.

3.3. Scope and Limitations

Scope and limitation 1: The development of the measure box and the collection of the dataset will be part of the work.

Scope and limitation 2: Only two ML approaches will be considered: CNN as the ground truth and ESN as a efficient solution.

Scope and limitation 3: The proposed system will not work on real time, even for the experimental evaluation the data will be first collected and later presented to the model in order to obtain the evaluation metrics.

Scope and limitation 3: The programming language will be python, the hardware to process the data will be a desktop personal computer.

Methodology

The following chapter details the physical setup to develop the measurements that constitute the dataset under study, also present a description of testbench and processing pipeline in which the data flows from the acoustic inputs towards the desire location-orientation output.

4.1. Experiment Design

The process starts with the data collection under an experimental setup, the data will be curated to eliminate random noise and presented as a time series dataset. The resulting dataset will be used to train and test four neural network pipelines. The predicted location-orientation error is calculated and a multifactor Analysis of Variance (ANOVA) is proposed as the main criteria to evaluate the significant variance of the error of each processing pipeline used to estimate the location and orientation of an object in a room.

4.2. Experimental measurement configuration

The research require the development of a manipulable and fully controllable environment in which the acoustic measurements are going to be performed. A small box type room is the volumetric cubic space used as the experimental container. The container used for the experimental measurements is a cubic box with sides of 1m in length and a volume close to $1m^3$, the concept is show in Figure 4.1 (a), three of its six walls are built from porcelain tiles with a thickness of 1cm, one wall an the floor are made of solid concrete, while the remaining wall used for observation is made of acrylic with a thickness of 0.5cm. The clear acrylic sheet allows direct observation of the movement mechanism inside the box. Unlike the other walls, which are permanently bonded, this acrylic wall can be removed to provide access to the interior of the measurement box. The movement mechanism inside the box consists of a robotic arm mounted at the center of the floor. The arm has four Degrees Of Freedom (DOF) and is used to move and change the position and orientation of a small cube with 20cm side length attached to its end. The cube is made of plastic, with walls of 2mm thickness. For the measurements, a mono speaker is placed at the center of one wall to generate the acoustic stimulus inside the system. On the opposite wall, four capacitive microphones are positioned 10cm away from each corner, with the purpose of capturing the acoustic response of the measurement box. The implementation of the measurement box was build inside a storage room and is presented in the Figure 4.1 (b).

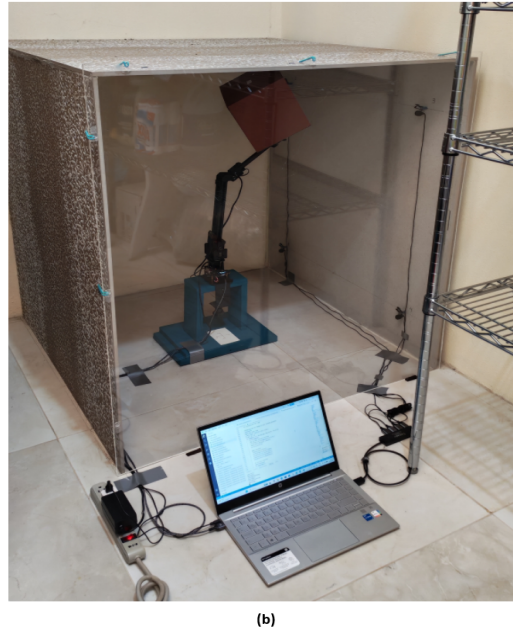
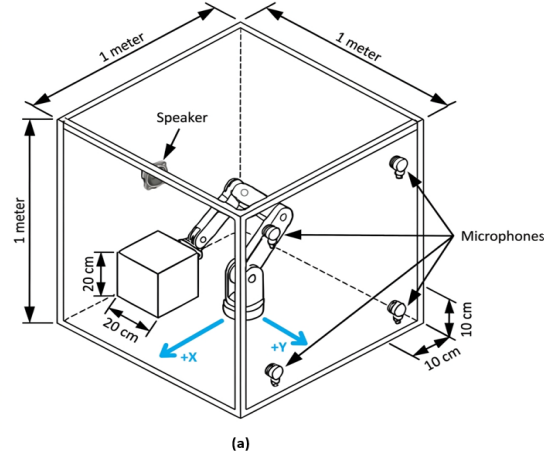


Figure 4.1: Measure box: (a) Conceptual sketch for the acoustic measurement box, and (b) actual implementation for the data acquisition task.

4.3. The Audio-Location Dataset for Model Training and Testing

The dataset was collected through twenty-six recording sessions. In each session, the robotic arm maintained a fixed angular value at its base and raised the cube to specific coordinates and orientations. The initial orientation of the cube was set with

its sides parallel to the measurement box walls. The variables modified during each measurement were the cube's location and orientation. For each location, nine different orientations were recorded. After completing these nine measurements, the robotic arm lowered the cube by decreasing its position along the Z axis, and the recording process was repeated for the new location. This procedure continued until the cube reached the bottom of the measurement box.

Audio from four channels was recorded. The acoustic signal was emitted four times, and the response was captured one microphone at a time. Each recording session consisted of 270 recordings, each containing four channels. A change of 10 degrees in the base of the robotic arm differentiated each recording session, producing a cylindrical envelope for the measurement area.

The coordinates of each measurement, obtained from the robotic arm, are shown in Fig. 4.2. The distribution of points is presented in the plot, each blue column in the three dimensions corresponds to the points in a recording session, and each point contains the nine orientations captured at that location. The total number of sessions was limited by the rotational constraint of the robotic arm's base, which allowed only 270 degrees of angular movement in its base. In total, 26 sessions were conducted, resulting in a dataset of 7020 recordings.

The source audio signal was obtained by arbitrarily extracting a 0.3-second segment from an acoustic melody. The spectrum of the signal has a bandwidth that reaches 10 kHz, with higher-magnitude components concentrated at low frequencies. Before reproducing the source signal, the recording process is initialized by opening the capture window for one second. This procedure is repeated four times, once for each microphone. After completing the four recordings, the cube is moved to a new orientation until it complete the nine possible and after this it moves to a new location.

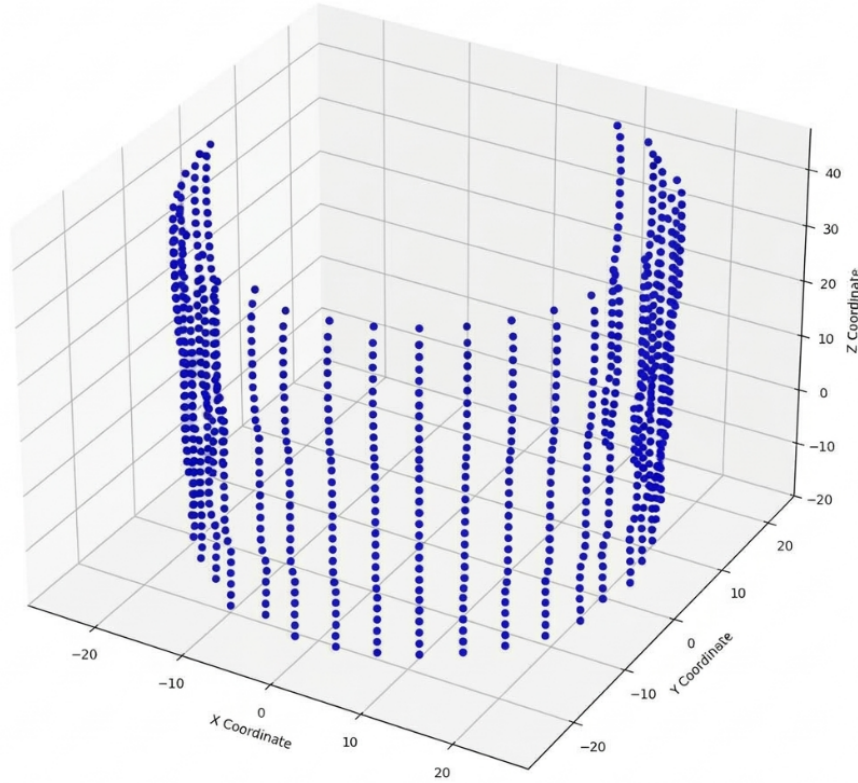


Figure 4.2: Scatter plot of cube locations for each recording in the dataset, the blue columns represent the 26 sessions, and each point in the column represents the nine orientations captured at that location.

4.4. The Audio-Location Processing Pipeline

In figure 4.3 can be observed the processing pipeline for the acoustics signals. From the measurement box, the microphone array records four signals to output four audio channels, these channels are used to feed the pipelines. First in AudAlign step the audio signals are aligned due to the temporal unpredictability in the recording start time caused by the OS process scheduler, the signals from the four channels are aligned by exploiting the fact that the distance from the source to each microphone is identical. Therefore, the magnitude of the direct sound can be used as an alignment reference for the signals. In the same step, a signal-integrity validation is performed in

which each recording is divided into its main body and tail (reverberation segment), from the main segment, the signal power is computed, and from the tail, its duration is extracted. Signals whose power and tail length fall within predefined limits are considered valid and included in the dataset. The resulting dataset is used in four different scenarios, two of them use a CNN-based pipeline and the other two use an ESN-based pipeline. The CNN-based require the time series data to be transformed into an image representation, while the ESN-based pipelines learn from an audio to spectral sequence series transformation. Additionally, a beamforming approach is suggested to address the superposition creating a segmented origin representation of the acoustic reverberation, dividing the room space into four different regions. The sound originating from these regions is contained in a group of newly created channels that will also feed both CNN and ESN estimators.

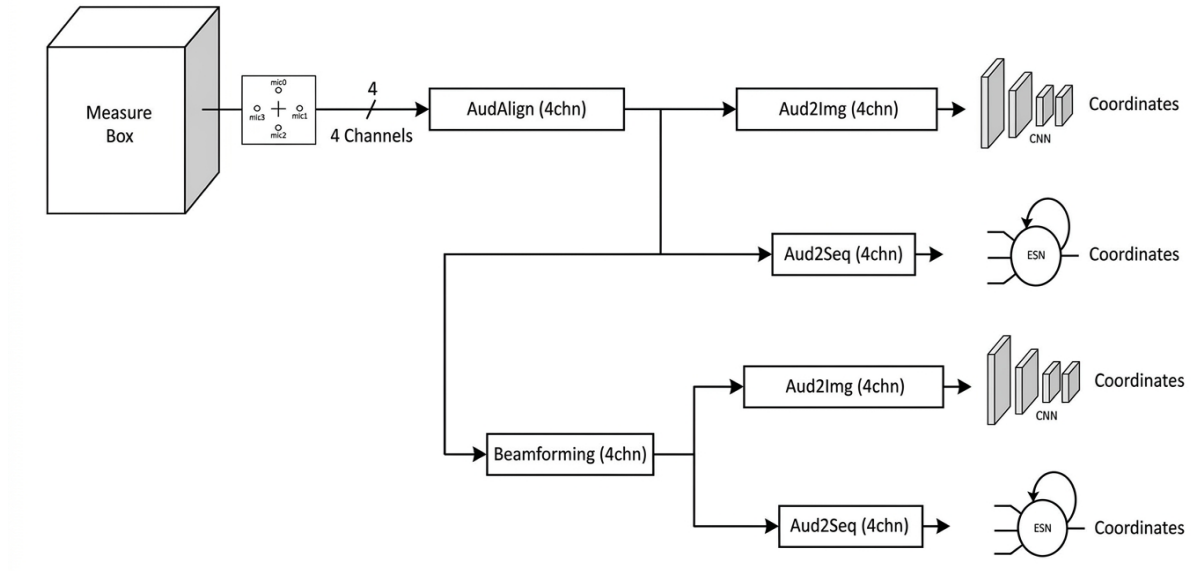


Figure 4.3: Acoustic to location processing pipeline

4.5. Factors and Levels

Factors are the independent variables that are going to be manipulated in the experiment, while levels are the different values that each factor can take. The factors and levels are selected based on the experimental design and the research questions being addressed. In this experiment, the factors include beamforming and network topology. Each factor has specific levels that will be tested to evaluate their impact in the task of estimating the location and orientation of the cube in the room.

For the experimental analysis, the factors and their respective levels to consider are summarized in the table 4.1. The network factor has two levels, The CNN is the reference model, the one that will be used as a baseline for comparison considering that once the audio signals are transformed into an image representation, the CNN is a well-established model for image processing tasks becoming a good candidate for the task of estimating the location and orientation of the cube in the room, the other level of the network factor is the ESN, which is an memory and training efficient model that will be evaluated against the CNN. Also a beamforming factor is considered, which has two levels, with and without beamforming. The combination of these factors and levels will allow for a comprehensive evaluation of the performance of the different processing pipelines in estimating the location and orientation of the cube in the room.

Table 4.1: Factors and levels for the experiment.

Levels	Factors	
	Network	Beamforming
	CNN ESN	With Without

4.6. Response Variables

The networks in this research are configured as regression models, the outputs are the predicted coordinates (X, Y, Z) and orientation of the cube in the room. The response variables are the error between the predicted and actual values of these outputs, which are calculated as mean of the absolute difference between the predicted and actual values as shows in equation 4.1. The Mean Absolute Error (MAE) is used as the performance metric for evaluating the models. "The MAE is a widely used metric for regression tasks, as it provides a clear measure of the average magnitude of errors in predictions, without considering their direction. Lower MAE values indicate better model performance, as they reflect smaller discrepancies between predicted and actual values".

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (4.1)$$

4.7. Data Recollection for Model Evaluation

To evaluate the performance of the models, a five-fold cross-validation approach are used. In this approach, the dataset is divided into five subsets, and the model is trained and tested five times, each time using a different subset as the test set and the remaining subsets as the training set. This process allows for a more robust evaluation of the model's performance and helps to mitigate any potential bias that may arise from using a single train-test split. The total 7020 samples are divided into five folds, with each fold containing near to 1,404 samples (A 80:20 ratio). The performance of the model is evaluated considering the Mean Absolute Error (MAE). The structure of the five-fold cross-validation and the corresponding performance metrics are summarized in Table 4.2. Due to the factor and levels we have 4 model variation to consider, and for each model variation the coordinates X, Y, Z, and orientation are evaluated, resulting

in a total of 20 MAE to obtain for evaluation.

Fold Iteration	Training Set	Test Set	Performance Metric
Fold 1	Fold 2, Fold 3, Fold 4, Fold 5	Fold 1	MAE ₁
Fold 2	Fold 1, Fold 3, Fold 4, Fold 5	Fold 2	MAE ₂
Fold 3	Fold 1, Fold 2, Fold 4, Fold 5	Fold 3	MAE ₃
Fold 4	Fold 1, Fold 2, Fold 3, Fold 5	Fold 4	MAE ₄
Fold 5	Fold 1, Fold 2, Fold 3, Fold 4	Fold 5	MAE ₅

Table 4.2: Five-fold cross-validation structure and performance summary.

4.8. Statistical Tool Used for hipotesis Testing

The groups of MAE obtained from the five-fold cross-validation for each model variation are considered as dependent samples because the same fold split of the dataset is used for each model evaluation. Two approaches can be followed to evaluate the significance of the differences in the MAE between the different model variations. The first approach is to use a repeated measures ANOVA, which is a statistical test that compares means across multiple related groups. in order to use this approach, the assumptions of normality and sphericity must be met. If these assumptions are not met, a non-parametric alternative such as the Friedman test can be used. The Friedman test is a non-parametric test that compares the ranks of the data across multiple related groups. It does not assume normality and is suitable for small sample sizes. In the present work The Friedman test is used as the recommended option to determine if there are significant statistical differences in the MAE between machine learning models [26].

In the friedman test, Instead of comparing the means of the groups, the Friedman test ranks the data within each fold. The ranks are then summed for each model variation, and the test statistic is calculated based on these rank sums. Finally The resulting values are compared to a chi-square distribution to find the p-value, which indicates the probability of observing the data if the null hypothesis is true [15].

The hypotheses for the test are as follows: H_0 : All models perform equally. H_1 : At

least one model's performance is significantly different.

If the $p - value < 0.05$, reject H_0 and a post-hoc analysis is required to determine which specific models differ from each other.

Chapter 5

Design

“Another quote.”

Another Author

This chapter presents the developed algorithm/mechanism used to test the hypothesis. The contents of this section vary depending on the research effort

5.1. Design criteria used (if needed)

5.1.1. Detailed specific criteria/tool

5.2. Description of the implementation of other algorithms (by other authors) used to test the hypothesis

5.3. Proposed algorithm

5.3.1. Design of the proposed algorithm

The specific description of the design algorithm goes in this place, as well as a flow diagram with the description of how it works as seen on **Figure 5.1**

5.3.2. Limitations and Requirements

A brief description of the design mechanism's limitations and requirements in accordance with the definitions in Scope and Limitations.

5.3.3. Implementation

A brief description of the implementation and pointer to git where it can be downloaded and tested. <https://myalgorithm.com>.

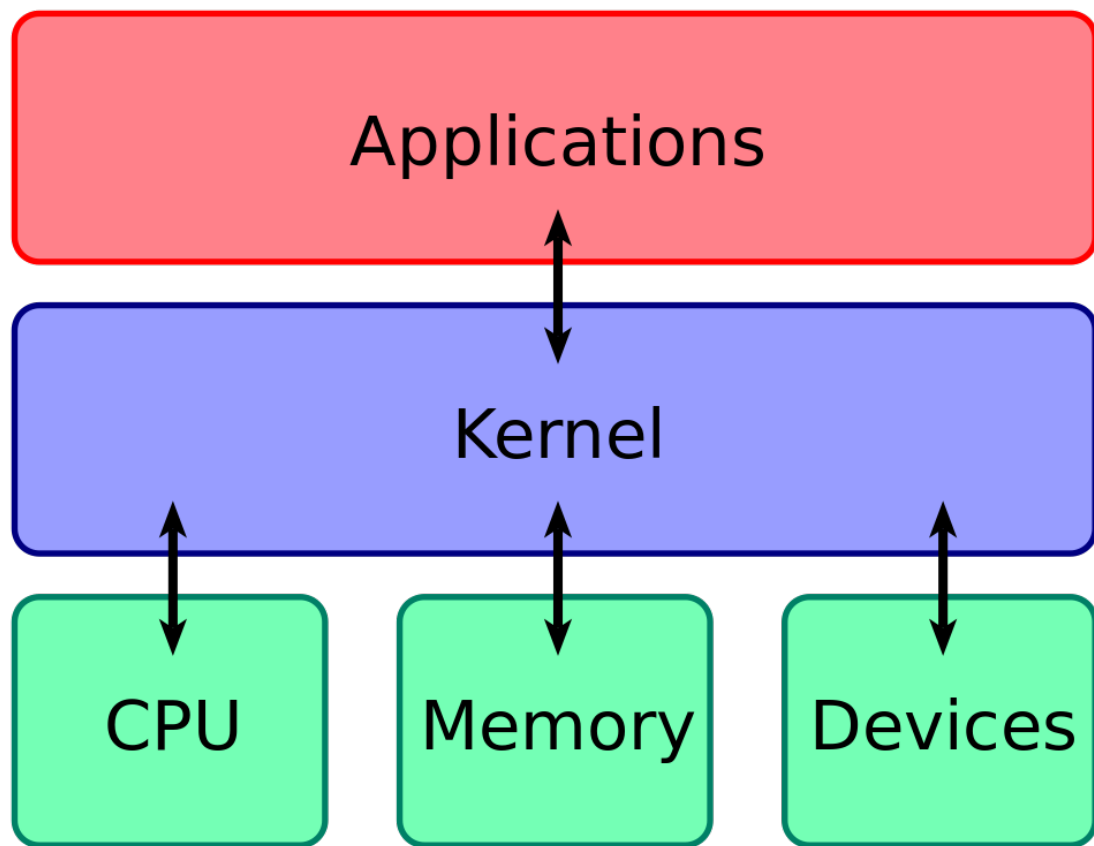


Figure 5.1: Example diagram.

Results

“Another quote.”

Another author

This chapter presents the results obtained while evaluating the proposed algorithm/mechanisms. The discussion and analysis of results are presented in the Discussion chapter.

6.1. Performance evaluation

6.2. Statistical Tool Assumptions Tests

The evaluation of the assumptions of the statistical tool used should be reported.

6.3. Statistical Tool Results

The results of the statistical tool used are presented in this section.

6.4. Obtained Metrics

Additional metrics obtained during the development of the research are presented in this section.

6.4.1. Most Complex Scenario Metrics (if needed)

6.5. Correctness (if needed)

If the correctness of the proposal was evaluated, it should be presented in this section.

6.6. Output (If needed)

If the proposal's output differs from what was analyzed with the statistical tool, it should be presented in this section.

Chapter 7

Discussion

“Another quote.”

Another Author

This chapter presents the analysis and discussion of the results shown in chapter Results.

Conclusions and Future Work

“Another quote.”

Another author

This chapter presents the final conclusions obtained through the research effort and the future work that can be derived.

8.1. Conclusions

- Conclusion 1.
- Conclusion 2.
- Conclusion 3.

8.2. Future Work

- Future work 1.
- Future work 2.
- Future work 3.

Chapter 9

Appendixes

9.1. Source Code

Pointer or description of the source code can be given in this section..

9.2. Gathered Data from the Experiments

If required, raw data could be presented in this section.

References

- [1] Jacob Bitterman, Daniel Levi, Hilel Hagai Diamandi, Sharon Gannot, and Tal Rosenwein. Revrir: Joint reverberant speech and room impulse response embedding using contrastive learning with application to room shape classification, 2024. URL <https://arxiv.org/abs/2406.03120>.
- [2] Amandine Brunetto, Sascha Hornauer, Stella X. Yu, and Fabien Moutarde. The audio-visual batvision dataset for research on sight and sound. In *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, page 1–8. IEEE, October 2023. doi: 10.1109/iros55552.2023.10341715. URL <http://dx.doi.org/10.1109/IROS55552.2023.10341715>.
- [3] Laurens Byttebier. *Small Footprint Neural Network Models for Acoustic Scene Classification*. Mtech thesis, Ghent University, 2021. Available at https://libstore.ugent.be/fulltxt/RUG01/003/014/483/RUG01-003014483_2021_0001_AC.pdf.
- [4] Ludovick Lepauloux Daniele Battaglino and Nicholas Evans. Acoustic scene classification using convolutional neural networks, 2016. URL https://dcase.community/documents/challenge2016/technical_reports/DCASE2016_Battaglino_1032.pdf.
- [5] Leandro de Santana. Fundamentals of acoustic beamforming, 2017. URL <https://www.sto.nato.int/publications/STO%20Educational%20Notes/STO-EN-AVT-287/EN-AVT-287-04.pdf>.
- [6] Hugo Demontis, Francois Ollivier, and Jacques Marchal. 3d identification of acoustic sources in rooms using a large-scale microphone array. In *2018 16th International Workshop on Acoustic Signal Enhancement (IWAENC)*, pages 506–509, 2018. doi: 10.1109/IWAENC.2018.8521264.
- [7] Maria Deprez and Emma C. Robinson. Chapter 11 - convolutional neural networks. In Maria Deprez and Emma C. Robinson, editors, *Machine Learning*

- for Biomedical Applications*, pages 233–270. Academic Press, 2024. ISBN 978-0-12-822904-0. doi: <https://doi.org/10.1016/B978-0-12-822904-0.00016-9>. URL <https://www.sciencedirect.com/science/article/pii/B9780128229040000169>.
- [8] Ivan Dokmanić, Reza Parhizkar, Andreas Walther, Yue M. Lu, and Martin Vetterli. Acoustic echoes reveal room shape. *Proceedings of the National Academy of Sciences*, 110(30):12186–12191, 2013. doi: 10.1073/pnas.1221464110. URL <https://www.pnas.org/doi/abs/10.1073/pnas.1221464110>.
- [9] Ivan Dokmanić, Reza Parhizkar, Andreas Walther, Yue M. Lu, and Martin Vetterli. Acoustic echoes reveal room shape. *Proceedings of the National Academy of Sciences*, 110(30):12186–12191, 2013. doi: 10.1073/pnas.1221464110. URL <https://www.pnas.org/doi/abs/10.1073/pnas.1221464110>.
- [10] Johann Faouzi. Time series classification: A review of algorithms and implementations. In Jorge Rocha, Cláudia M. Viana, and Sandra Oliveira, editors, *Time Series Analysis*, chapter 2. IntechOpen, Rijeka, 2024. doi: 10.5772/intechopen.1004810. URL <https://doi.org/10.5772/intechopen.1004810>.
- [11] Òscar Garibo-i Orts, Nicolas Firbas, Laura Sebastiá, and J. Alberto Conejero. Gramian angular fields for leveraging pretrained computer vision models with anomalous diffusion trajectories. *Phys. Rev. E*, 107:034138, Mar 2023. doi: 10.1103/PhysRevE.107.034138. URL <https://link.aps.org/doi/10.1103/PhysRevE.107.034138>.
- [12] Letizia Gionfrida, Chen Wang, Lu Gan, Margarita Chli, and Luca Carlone. Computer and robot vision: Past, present, and future [tc spotlight]. *IEEE Robotics & Automation Magazine*, 31(3):211–215, 2024. doi: 10.1109/MRA.2024.3428780.
- [13] Yoonchang Han, Jeongsoon Park, and Kyogu Lee. Convolutional neural networks with binaural representations and background subtraction for acoustic scene classification. In *Workshop on Detection and Classification of Acoustic Scenes and Events*, 2017. URL <https://api.semanticscholar.org/CorpusID:52830611>.
- [14] Gunnar Heilmann, Andy Meyer, and Dirk Döbler. Beamforming in the time-domain using 3d-microphone arrays. In *XIXth Biennial Conference of the New Zealand Acoustical Society, Auckland*, 2008.
- [15] Julien I.E. Hoffman. Chapter 26 - analysis of variance. ii. more complex forms. In *Basic Biostatistics for Medical and Biomedical Practitioners (Second Edition)*, pages 419–441. Academic Press, second edition edition, 2019. ISBN 978-0-12-817084-7. doi: <https://doi.org/10.1016/B978-0-12-817084-7.00026-7>. URL <https://www.sciencedirect.com/science/article/pii/B9780128170847000267>.
- [16] Shengli Jiang, Shiyi Qin, Joshua L. Pulsipher, and Victor M. Zavala. Chapter 3 - convolutional neural networks: Basic concepts and applications in manufacturing. In Masoud Soroush and Richard D Braatz, editors, *Artificial Intelligence in Manufacturing*, pages 63–102. Academic Press, 2024. ISBN 978-0-323-99134-6.

- doi: <https://doi.org/10.1016/B978-0-323-99134-6.00007-4>. URL <https://www.sciencedirect.com/science/article/pii/B9780323991346000074>.
- [17] Shaowei Jin, Huaping Liu, Bowen Wang, and Fuchun Sun. Open-environment robotic acoustic perception for object recognition. *Frontiers in Neurorobotics*, Volume 13 - 2019, 2019. ISSN 1662-5218. doi: 10.3389/fnbot.2019.00096. URL <https://www.frontiersin.org/journals/neurorobotics/articles/10.3389/fnbot.2019.00096>.
- [18] Jacqueline Koeppen and Gunnar Heilmann. 3d beam forming measurements using 3d-microphone arrays. Technical report, SAE Technical Paper, 2009.
- [19] Teodoro Aguilera Leticia Jaén, Fernando J. Álvarez and Juan J. García. Groom-level indoor positioning based on acoustic impulse response identification. *International Conference on Indoor Positioning and Indoor Navigation*, September 2017.
- [20] Mantas Lukoševičius and Herbert Jaeger. Reservoir computing approaches to recurrent neural network training. *Computer Science Review*, 3(3):127–149, 2009. ISSN 1574-0137. doi: <https://doi.org/10.1016/j.cosrev.2009.03.005>. URL <https://www.sciencedirect.com/science/article/pii/S1574013709000173>.
- [21] Shan Luo, Leqi Zhu, Kaspar Althoefer, and Hongbin Liu. Knock-knock: Acoustic object recognition by using stacked denoising autoencoders. *Neurocomputing*, 267:18–24, 2017. ISSN 0925-2312. doi: <https://doi.org/10.1016/j.neucom.2017.03.014>. URL <https://www.sciencedirect.com/science/article/pii/S092523121730509X>.
- [22] Tomoya Manabe, Rikuto Fukunaga, Kei Nakatsuma, and Makoto Kumon. Object surface recognition using microphone array by acoustic standing wave. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 9274–9279, 2022. doi: 10.1109/IROS47612.2022.9981386.
- [23] Iain McCowan. Microphone arrays : A tutorial. In *Computer Science, Engineering*, 2001. URL <https://api.semanticscholar.org/CorpusID:16623291>.
- [24] mordorintelligence. Usa maintenance repair & operations industry size & share analysis - growth trends & forecasts (2025 - 2030), 2025. URL <https://www.mordorintelligence.com/industry-reports/united-states-maintenance-repair-and-operations-mro-market>.
- [25] Nils Peters, Howard Lei, and Gerald Friedland. Name that room: room identification using acoustic features in a recording. MM '12, page 841–844, New York, NY, USA, 2012. Association for Computing Machinery. ISBN 9781450310895. doi: 10.1145/2393347.2396326. URL <https://doi.org/10.1145/2393347.2396326>.

- [26] Mauri Rainio, Aleksi Teuho, and Riku Klén. Evaluation metrics and statistical tests for machine learning. *Scientific Reports*, 14:6038, 2024. doi: 10.1038/s41598-024-56706-x. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC10937649/>.
- [27] Michael Rothstein, Zhuolin Liu, and Islam Bashima. Echosense: Acoustic sensing for enclosed structure monitoring and recognition. In *2023 IEEE MIT Undergraduate Research Technology Conference (URTC)*, pages 1–6, 2023. doi: 10.1109/URTC60662.2023.10534944.
- [28] Faizan Samad. Using audio ray tracing to calculate room impulse response, 2009. URL [https://courses.washington.edu/phys536/presentations/mar%2009/samad-Applications%20of%20Audio%20Ray%20Tracing%20\(1\).pdf](https://courses.washington.edu/phys536/presentations/mar%2009/samad-Applications%20of%20Audio%20Ray%20Tracing%20(1).pdf).
- [29] CAE Software, Systems, and 01db. Noise inspector - acoustic cameras technical datasheet, 2016. URL <https://www.gkmconsultants.com/assets/documents/instruments/pdf/not1542-june-2016-e-noise-inspector-acoustic-camera-technical-data-sheet-en.pdf>.
- [30] Jasna Stamać, Sanja Grubeša, and Antonio Petošić. Designing the acoustic camera using matlab with respect to different types of microphone arrays. In *2019 2nd International Colloquium on Smart Grid Metrology (SMAGRIMET)*, pages 1–6, 2019. doi: 10.23919/SMAGRIMET.2019.8720363.
- [31] Winter 2007 stanford edu, Music 318. Room impulse response measurement and analysis, 2007. URL <https://ccrma.stanford.edu/~adnanm/SCI220/Music318ir.pdf>.
- [32] Chenxi Sun, Moxian Song, Shenda Hong, and Hongyan Li. A review of designs and applications of echo state networks. *CoRR*, abs/2012.02974, 2020. URL <https://arxiv.org/abs/2012.02974>.
- [33] Iuliana Tabian, Hailing Fu, and Zahra Sharif Khodaei. A convolutional neural network for impact detection and characterization of complex composite structures. *Sensors*, 19(22), 2019. ISSN 1424-8220. doi: 10.3390/s19224933. URL <https://www.mdpi.com/1424-8220/19/22/4933>.
- [34] Ethan Tracy and Navinda Kottege. Catchatter: Acoustic perception for mobile robots. *IEEE Robotics and Automation Letters*, 6(4):7209–7216, 2021. doi: 10.1109/LRA.2021.3094492.
- [35] Van-Thuan Tran, Chia-Yang Wu, and Wei-Ho Tsai. Acoustic-based lego recognition using attention-based convolutional neural networks. *Artif. Intell. Rev.*, 57(1), January 2024. ISSN 0269-2821. doi: 10.1007/s10462-023-10625-x. URL <https://doi.org/10.1007/s10462-023-10625-x>.

- [36] Spoorthy. V, Manjunath Mulimani, and Shashidhar G. Koolagudi. Acoustic scene classification using deep learning architectures. In *2021 6th International Conference for Convergence in Technology (I2CT)*, pages 1–6, 2021. doi: 10.1109/I2CT51068.2021.9418177.
- [37] Víctor Manuel Vargas, Rafael Ayllón-Gavilán, Antonio Manuel Durán-Rosal, Pedro Antonio Gutiérrez, César Hervás-Martínez, and David Guijo-Rubio. Gramian angular and markov transition fields applied to time series ordinal classification. In Ignacio Rojas, Gonzalo Joya, and Andreu Catala, editors, *Advances in Computational Intelligence*, pages 505–516, Cham, 2023. Springer Nature Switzerland. ISBN 978-3-031-43078-7.
- [38] Andrea Vitali. Microphone array beamforming in the pcm and pdm domain, 2018. URL https://www.st.com/resource/en/design_tip/dt0117-microphone-array-beamforming-in-the-pcm-and-pdm-domain-stmicroelectronics.pdf.
- [39] Mason Wang, Samuel Clarke, Jui-Hsien Wang, Ruohan Gao, and Jiajun Wu. Soundcam: A dataset for finding humans using room acoustics, 2024. URL <https://arxiv.org/abs/2311.03517>.
- [40] BBC wildlife. What exactly is echolocation and how does it work? the incredible adaptation that allows animals to ‘see’ and communicate in the dark, 2025. URL <https://www.discoverwildlife.com/animal-facts/what-is-echolocation>.
- [41] Junhyuk Woo, Hyeongmo Kim, Soon Ho Kim, and Kyungreem Han. Echo state property upon noisy driving input. *Complexity*, 2024(1):5593925, 2024. doi: <https://doi.org/10.1155/2024/5593925>. URL <https://onlinelibrary.wiley.com/doi/abs/10.1155/2024/5593925>.
- [42] Rikiya Yamashita, , Rikiya Yamashita, Richard Kinh Gian, and Kaori Togashi. Convolutional neural networks: an overview and application in radiology. In *nsights Imaging*, page 611–629. springeropen, 2018. ISBN 978-0-12-822904-0. doi: <https://doi.org/10.1007/s13244-018-0639-9>. URL <https://insightsimaging.springeropen.com/articles/10.1007/s13244-018-0639-9#citeas>.
- [43] Zhoutong Zhang. *Inferring Shape and Material from Sound*. Mtech thesis, MIT, 2021. Available at <https://dspace.mit.edu/handle/1721.1/139579>.